Math 401 — Applications of Linear Algebra

Course Notes

Least Squares · Curve Fitting · Team Ranking · Graph Theory · Markov Chains · PageRank · SVD · Image Compression · Quantum Qubit

Based on notes by Justin Wyss-Gallifent
and Lay & McDonald, *Linear Algebra and Its Applications*, Ch. 10

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Chapter 1

Linear Algebra Review

This chapter summarizes the prerequisite linear algebra needed for this course. Source: Justin Wyss-Gallifent, *Math 401/431: Linear Algebra Review.*

1.1 Matrices and Vectors

Definition 1.1.1 (Matrix)**.** A **matrix** is a rectangular array of numbers. An $n \times m$ matrix has n rows and m columns. A **vector** is an $n \times 1$ matrix.

Definition 1.1.2 (Matrix–Vector Product)**.** If A is $n \times m$ and $\mathbf{b}$ is $m \times 1$, the product $A\mathbf{b}$ is the linear combination of columns of A weighted by the entries of $\mathbf{b}$:

$$A\mathbf{b} = b_1\mathbf{a}_1 + \cdots + b_m\mathbf{a}_m.$$

Definition 1.1.3 (Transpose, Symmetry, Inverse)**.**

- The **transpose** A^T is obtained by swapping rows and columns: $(A^T)_{ij} = A_{ji}$.
- A is **symmetric** if $A^T = A$.
- A square matrix A is **invertible** if there exists A^{-1} with $AA^{-1} = A^{-1}A = I$.

Theorem 1.1.1 (2×2 Inverse)**.** $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \dfrac{1}{ad - bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ *provided* $ad - bc \neq 0$.

Definition 1.1.4 (Dot Product and Norm)**.** For $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$:

$$\mathbf{v} \cdot \mathbf{w} = \mathbf{v}^T\mathbf{w} = v_1w_1 + \cdots + v_nw_n, \qquad \|\mathbf{v}\| = \sqrt{v_1^2 + \cdots + v_n^2}.$$

1.2 Determinants

Definition 1.2.1 (Determinant)**.** For a 2×2 matrix: $\det\begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$. For larger matrices, expand along the first row:

$$\det(A) = a_{11}\det(A_{11}) - a_{12}\det(A_{12}) + \cdots \pm a_{1n}\det(A_{1n}),$$

where A_{1j} is the matrix with row 1 and column j removed.

Theorem 1.2.1. *A square matrix A is invertible if and only if* $\det(A) \neq 0$.

1.3 Systems of Equations

Theorem 1.3.1. *Any linear system $a_{i1}x_1 + \cdots + a_{in}x_n = b_i$ can be written as $A\mathbf{x} = \mathbf{b}$. It has:*

- *Exactly one solution if A is invertible (then $\mathbf{x} = A^{-1}\mathbf{b}$).*
- *No solution or infinitely many solutions otherwise.*

1.4 Linear Independence

Definition 1.4.1. A set $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is **linearly independent** if $a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n = \mathbf{0}$ implies $a_1 = \cdots = a_n = 0$. Otherwise it is **linearly dependent** (one vector is a linear combination of the others).

1.5 Vector Spaces and Bases

Definition 1.5.1 (Vector Space)**.** A nonempty set V of vectors is a **vector space** (subspace of $\mathbb{R}^n$) if it contains $\mathbf{0}$, is closed under addition, and is closed under scalar multiplication.

Definition 1.5.2 (Span, Basis, Column Space)**.**

- $\text{span}(S) = \{a_1\mathbf{v}_1 + \cdots + a_k\mathbf{v}_k \mid a_i \in \mathbb{R}\}$ for $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$.
- A **basis** for V is a linearly independent set B with $\text{span}(B) = V$.
- The **column space** of A is $\text{Col}(A) = \text{span}(\text{columns of } A) = \{A\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^m\}$.
- The **dimension** $\dim(V)$ is the number of vectors in any basis of V.

1.6 Orthogonality and Orthonormality

Definition 1.6.1 (Orthogonal and Orthonormal Sets)**.**

- $\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ is **orthogonal** if $\mathbf{v}_i \cdot \mathbf{v}_j = 0$ for $i \neq j$.
- It is **orthonormal** if additionally $\|\mathbf{v}_i\| = 1$ for all i.

Definition 1.6.2 (Orthogonal Matrix)**.** A square matrix is **orthogonal** if its columns form an orthonormal set, equivalently $A^TA = AA^T = I$, i.e., $A^{-1} = A^T$.

Remark 1.6.1. Orthogonal matrices are computationally convenient because $A^{-1} = A^T$ is trivially obtained. They arise naturally in QR decompositions used for least-squares computation.

1.7 Eigenvalues and Eigenvectors

Definition 1.7.1 (Eigenvalue and Eigenvector)**.** If A is $n \times n$, then λ is an **eigenvalue** and $\mathbf{v} \neq \mathbf{0}$ is the corresponding **eigenvector** if $A\mathbf{v} = \lambda\mathbf{v}$.

Definition 1.7.2 (Characteristic Polynomial)**.** $\text{char}(A) = \det(\lambda I - A)$. The eigenvalues of A are the roots of $\text{char}(A)$.

Fact 1.7.1. *For a 2×2 matrix:* $\text{char}(A) = \lambda^2 - \text{tr}(A)\lambda + \det(A)$. *For a triangular matrix: eigenvalues are the diagonal entries.*

Example 1.7.1. $A = \begin{pmatrix} 4 & 7 \\ 1 & -2 \end{pmatrix}$. Then $\text{char}(A) = \lambda^2 - 2\lambda - 15 = (\lambda - 5)(\lambda + 3)$. Eigenvalues: $\lambda_1 = 5$ with eigenvector $\begin{pmatrix} 7 \\ 1 \end{pmatrix}$; $\lambda_2 = -3$ with eigenvector $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

Theorem 1.7.1. *The eigenspace for eigenvalue λ (the set of all eigenvectors plus $\mathbf{0}$) is a vector space with dimension $\leq$ multiplicity of λ as a root of* $\text{char}(A)$.

1.8 Diagonalizable Matrices

Definition 1.8.1. A is **diagonalizable** if $A = PDP^{-1}$ for invertible P and diagonal D. The columns of P are eigenvectors and the diagonal of D contains corresponding eigenvalues.

Theorem 1.8.1. *A is diagonalizable $\iff$ the dimension of each eigenspace equals the multiplicity of the eigenvalue.*

Theorem 1.8.2 (Orthogonal Diagonalization)**.** *A is **orthogonally diagonalizable** (i.e., $A = QDQ^T$ with Q orthogonal) if and only if A is symmetric.*

Remark 1.8.1. Diagonalization is useful for computing powers: $A^k = PD^kP^{-1}$. Since D is diagonal, $D^k = \text{diag}(\lambda_1^k, \ldots, \lambda_n^k)$ is easy to compute. This is fundamental to analyzing Markov chain convergence.

1.9 Linear Transformations

Definition 1.9.1. $T : \mathbb{R}^n \to \mathbb{R}^m$ is **linear** if $T(\alpha\mathbf{v} + \beta\mathbf{w}) = \alpha T(\mathbf{v}) + \beta T(\mathbf{w})$ for all scalars α, β and vectors $\mathbf{v}, \mathbf{w}$.

Theorem 1.9.1. *Every linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is represented by an $m \times n$ matrix $A = [T(\mathbf{e}_1) \ \cdots \ T(\mathbf{e}_n)]$, so $T(\mathbf{v}) = A\mathbf{v}$ for all $\mathbf{v}$.*

Example 1.9.1. $T(x_1, x_2) = (2x_1,\, x_2 - x_1,\, 3x_1)$ is linear with matrix $A = \begin{pmatrix} 2 & 0 \\ -1 & 1 \\ 3 & 0 \end{pmatrix}$.

Note: $T(\mathbf{0}) = \mathbf{0}$ is a necessary condition for linearity. If $T(\mathbf{0}) \neq \mathbf{0}$ (e.g., $T(x) = x + 2$), the transformation is not linear.

Chapter 2

Least Squares

2.1 Motivation

We study the matrix equation $A\mathbf{x} = \mathbf{b}$. When no exact solution exists (i.e., $\mathbf{b} \notin \mathrm{Col}(A)$), we seek the best approximate solution – the one that makes $A\hat{\mathbf{x}}$ as close as possible to $\mathbf{b}$.

2.2 The Normal Equation

Geometric Idea

The vector in $\mathrm{Col}(A)$ closest to $\mathbf{b}$ is the orthogonal projection $\hat{\mathbf{b}} = \mathrm{Pr}_{\mathrm{Col}(A)}\,\mathbf{b}$. We then seek $\hat{\mathbf{x}}$ such that $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$.

Derivation

Since $\hat{\mathbf{b}} - \mathbf{b} \perp \mathrm{Col}(A)$, for every vector $A\bar{x} \in \mathrm{Col}(A)$:

$$(A\bar{x}) \cdot (A\hat{\mathbf{x}} - \mathbf{b}) = 0 \quad \text{for all } \bar{x} \implies A^T(A\hat{\mathbf{x}} - \mathbf{b}) = \mathbf{0}.$$

This gives the **normal equation**:

$$\boxed{A^T A\hat{\mathbf{x}} = A^T\mathbf{b}.}$$

Pythagorean Justification (Why This Minimizes Error)

For any $\bar{x}$, write $A\bar{x} - \mathbf{b} = (A\hat{\mathbf{x}} - \mathbf{b}) + A(\bar{x} - \hat{\mathbf{x}})$. Since these two terms are perpendicular:

$$\|A\bar{x} - \mathbf{b}\|^2 = \|A\hat{\mathbf{x}} - \mathbf{b}\|^2 + \|A(\bar{x} - \hat{\mathbf{x}})\|^2 \geq \|A\hat{\mathbf{x}} - \mathbf{b}\|^2.$$

2.3 Definitions and Main Theorem

Definition 2.3.1 (Least-Squares Solution)**.** A **least-squares solution** to $A\mathbf{x} = \mathbf{b}$ is any $\hat{\mathbf{x}}$ satisfying $\|A\hat{\mathbf{x}} - \mathbf{b}\| \leq \|A\bar{x} - \mathbf{b}\|$ for all $\bar{x}$.

Definition 2.3.2 (Least-Squares Error)**.** The **least-squares error** is $\|A\hat{\mathbf{x}} - \mathbf{b}\|$.

Theorem 2.3.1 (Uniqueness)**.**

- *If the columns of A are **linearly independent**, the least-squares solution is unique: $\hat{\mathbf{x}} = (A^T A)^{-1} A^T\mathbf{b}$.*
- *If the columns are **linearly dependent**, there are infinitely many least-squares solutions.*

Corollary 2.3.2. *If A is invertible, then $(A^T A)^{-1} A^T\mathbf{b} = A^{-1}(A^T)^{-1} A^T\mathbf{b} = A^{-1}\mathbf{b}$, i.e., least squares recovers the exact solution.*

Fact 2.3.1. *The normal equation always has at least one solution (since $A^T\mathbf{b} \in \mathrm{Col}(A^TA)$). This is why a least-squares solution always exists.*

2.4 Method: Orthogonal Projection via Gram-Schmidt

An alternative to the normal equation is to find an orthogonal basis for $\mathrm{Col}(A)$ using Gram-Schmidt, then project $\mathbf{b}$ directly:

Gram-Schmidt: Given linearly independent $\mathbf{c}_1, \mathbf{c}_2$:

$$\mathbf{c}_2^\perp = \mathbf{c}_2 - \frac{\mathbf{c}_1 \cdot \mathbf{c}_2}{\mathbf{c}_1 \cdot \mathbf{c}_1}\mathbf{c}_1.$$

Projection:

$$\mathrm{Pr}_{\mathrm{Col}(A)}\,\mathbf{b} = \frac{\mathbf{b}\cdot\mathbf{c}_1}{\|\mathbf{c}_1\|^2}\mathbf{c}_1 + \frac{\mathbf{b}\cdot\mathbf{c}_2^\perp}{\|\mathbf{c}_2^\perp\|^2}\mathbf{c}_2^\perp.$$

2.5 Worked Examples

Example 2.5.1 (Unique Least-Squares Solution)**.** Find $\hat{\mathbf{x}}$ for $\begin{pmatrix}1 & 2\\ 1 & 1\\ 1 & -1\end{pmatrix}\mathbf{x} = \begin{pmatrix}6\\4\\1\end{pmatrix}$.

Normal equations:

$$A^TA = \begin{pmatrix}3 & 2\\ 2 & 6\end{pmatrix}, \quad A^T\mathbf{b} = \begin{pmatrix}11\\15\end{pmatrix}.$$

$\det(A^TA) = 14$. Then:

$$\hat{\mathbf{x}} = \frac{1}{14}\begin{pmatrix}6 & -2\\ -2 & 3\end{pmatrix}\begin{pmatrix}11\\15\end{pmatrix} = \begin{pmatrix}36/14\\23/14\end{pmatrix} = \begin{pmatrix}18/7\\23/14\end{pmatrix}.$$

Error: $\|A\hat{\mathbf{x}} - \mathbf{b}\| \approx 0.267$.

Example 2.5.2 (Closest Point on a Line)**.** Find the point on $y = 3x$ closest to $(2, 3)$.

Matrix form: $\begin{pmatrix}1\\3\end{pmatrix} t = \begin{pmatrix}2\\3\end{pmatrix}$.

Normal equation: $10\hat{t} = 11 \implies \hat{t} = \frac{11}{10} = 1.1$.

Closest point: $(1.1,\ 3.3)$.

Example 2.5.3 (Closest Point on a Plane)**.** Find the point on the plane $2x + 3y = z$ closest to $\mathbf{b} = (1, 0, 0)$.

The plane has normal $(2, 3, -1)$, but as a subspace $\{(x, y, 2x + 3y)\}$, it is spanned by $\mathbf{c}_1 = (1, 0, 2)$ and $\mathbf{c}_2 = (0, 1, 3)$.

$A = \begin{pmatrix}1 & 0\\ 0 & 1\\ 2 & 3\end{pmatrix}$. Normal equations: $A^TA = \begin{pmatrix}5 & 6\\ 6 & 10\end{pmatrix}$, $A^T\mathbf{b} = \begin{pmatrix}1\\0\end{pmatrix}$.

$\det(A^TA) = 14$. $\hat{\mathbf{x}} = \frac{1}{14}\begin{pmatrix}10 & -6\\ -6 & 5\end{pmatrix}\begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}5/7\\-3/7\end{pmatrix}$.

Closest point: $A\hat{\mathbf{x}} = \begin{pmatrix}5/7\\-3/7\\-1/7\end{pmatrix}$.

Example 2.5.4 (Infinitely Many Solutions)**.** $\begin{pmatrix}1 & 2\\ 1 & 2\end{pmatrix}\mathbf{x} = \begin{pmatrix}4\\3\end{pmatrix}$. The columns are dependent, so $A^TA = \begin{pmatrix}2 & 4\\ 4 & 8\end{pmatrix}$ is singular. Solutions: $\hat{\mathbf{x}} = \begin{pmatrix}7/2 - 2t\\ t\end{pmatrix}$ for $t \in \mathbb{R}$. All give the same projection (and error).

2.6 Common Misconceptions

Remark 2.6.1 (Why $(A^TA)^{-1}A^T \neq A^{-1}$). Writing $(A^TA)^{-1}A^T = A^{-1}(A^T)^{-1}A^T = A^{-1}$ is **wrong** unless A is square and invertible. In the least-squares setting, A is typically $m \times n$ with $m > n$ (not square), so A^{-1} and $(A^T)^{-1}$ do not exist.

Remark 2.6.2 (Switching Columns). Switching columns of A (multiplying by a permutation P: $A' = AP$) does not change $\mathrm{Col}(A)$ or the minimum error. The solution changes as $\hat{\mathbf{x}}' = P^T\hat{\mathbf{x}}$ (just reorders components).

Chapter 3

Curve Fitting

3.1 Least-Squares Line

Given points $(x_1, y_1), \ldots, (x_n, y_n)$, fit $f(x) = mx + b$. Writing one equation per point:

$$\underbrace{\begin{pmatrix} x_1 & 1 \\ \vdots & \vdots \\ x_n & 1 \end{pmatrix}}_{A} \begin{pmatrix} m \\ b \end{pmatrix} = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \mathbf{b}.$$

Solve the normal equation $A^T A\hat{\mathbf{c}} = A^T\mathbf{b}$. This minimizes $\sum(y_i - f(x_i))^2$ (sum of squared *vertical* distances).

Remark 3.1.1. The line $y = mx+b$ and the line $x = ny+c$ (which minimizes horizontal distances) are generally **different** because they minimize different objective functions.

3.2 General Function Fitting: What Works?

Least squares applies whenever the unknown coefficients appear **linearly**:

$$f(x) = a_1 f_1(x) + \cdots + a_k f_k(x),$$

where $f_1, \ldots, f_k$ are known functions. The matrix is:

$$A = \begin{pmatrix} f_1(x_1) & \cdots & f_k(x_1) \\ \vdots & & \vdots \\ f_1(x_n) & \cdots & f_k(x_n) \end{pmatrix}.$$

Example 3.2.1 (What Works and What Doesn't)**.** Matrix column i: (e^{x_i}, x_i).

- $f(x) = ae^x + bx$: **Yes** (linear in a, b).
- $f(x) = a\sin x + b\cos x + c$: **Yes** (linear in a, b, c).
- $f(x) = e^{ax} + bx$: **No** (a appears in exponent).
- $f(x) = a\sin(bx) + c$: **No** (b appears inside sine).
- $f(x) = ax^2 + bx^c$: **No** (c appears as exponent).

3.3 Common Fitting Templates

Polynomial Fit

Parabola $f(x) = ax^2 + bx + c$: column i of A is $(x_i^2, x_i, 1)$.
Degree-k polynomial: column i of A is $(x_i^k, x_i^{k-1}, \ldots, x_i, 1)$.

Exponential Fit

$f(x) = ae^x + b$: column i of A is $(e^{x_i}, 1)$.

Sinusoidal Fit

$f(x) = a + b\sin x$: column i of A is $(1, \sin x_i)$.

Surface Fitting

For $f(x, y) = a_1 g_1(x, y) + \cdots + a_k g_k(x, y)$, each data point (x_i, y_i, z_i) gives one row. Example: paraboloid $f(x, y) = ax^2 + by^2$, matrix row i: (x_i^2, y_i^2).

Ellipse Fitting (Polar Form)

For an ellipse $Ar^2\cos^2\theta + Br^2\sin^2\theta = 1$, rewrite as:

$$(r_i\cos\theta_i)^2 \cdot A + (r_i\sin\theta_i)^2 \cdot B = 1.$$

Matrix row i: $(r_i^2\cos^2\theta_i,\ r_i^2\sin^2\theta_i)$, right-hand side all 1s.

3.4 Worked Examples

Example 3.4.1 (Line Through Three Points)**.** Points $(-1, 0), (1, 2), (2, 2)$.

$$A = \begin{pmatrix} -1 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}.$$

$A^T A = \begin{pmatrix} 6 & 2 \\ 2 & 3 \end{pmatrix}$, $A^T\mathbf{b} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$, $\det = 14$.

$$\hat{\mathbf{c}} = \frac{1}{14}\begin{pmatrix} 3 & -2 \\ -2 & 6 \end{pmatrix}\begin{pmatrix} 5 \\ 4 \end{pmatrix} = \begin{pmatrix} 1/2 \\ 11/7 \end{pmatrix}.$$

Line: $f(x) = \frac{1}{2}x + \frac{11}{7} \approx 0.5x + 1.571$.

Example 3.4.2 (Parabola Fit)**.** Points $(-1, 3), (0, 1), (1, 2), (3, 9)$. Set up A with columns $(x_i^2, x_i, 1)$ and solve. The least-squares parabola is approximately $f(x) \approx 1.045x^2 - 0.555x + 1.291$.

Example 3.4.3 (Sinusoidal Fit)**.** Points $(-2, 6.3), (3, 1.2), (5, 7.1), (8, -2.8), (9, -0.05)$. Fit $f(x) = a + b\sin x$.

$$A = \begin{pmatrix} 1 & \sin(-2) \\ 1 & \sin 3 \\ 1 & \sin 5 \\ 1 & \sin 8 \\ 1 & \sin 9 \end{pmatrix}.$$

Solving: $f(x) \approx 2.026 - 4.974\sin x$. Predict $f(\pi/2) \approx -2.948$.

3.5 Using the Best-Fit Function

After fitting:

1. Evaluate f at any point to make predictions.
2. Solve $f(x) = c$ to find when the function reaches a target.
3. Compute f' to find rates of change.
4. Compare the least-squares errors of competing models to select the best fit.

Chapter 4

Team Ranking: Massey's Method

4.1 Setup

Given T teams and game outcomes (each described by a winning margin), assign rankings $r_1, \ldots, r_T$.

Key premise: If team i beats team j by p points, the rankings should satisfy $r_i - r_j = p$.

For each game, write one equation. This gives $A\mathbf{r} = \mathbf{p}$ where each row of A has $+1$ in column i, -1 in column j, and 0 elsewhere.

Since game results are often inconsistent, $A\mathbf{r} = \mathbf{p}$ rarely has an exact solution. We use least squares.

4.2 Massey's Method

The Massey Matrix

The least-squares normal equation is $M\hat{\mathbf{r}} = \bar{q}$ where $M = A^T A$ and $\bar{q} = A^T \mathbf{p}$.

Structure of M:

- m_{ii} = total games team i played.
- $m_{ij} = -($ number of games between teams i and $j)$ for $i \neq j$.
- M is symmetric. Its rows all sum to zero $\Rightarrow$ M is singular.

Structure of $\bar{q}$:

- q_i = sum of all point margins for team i (wins positive, losses negative).
- Note: $\sum_i q_i = 0$ always (each game contributes $+p$ to one team and $-p$ to the other).

Fact 4.2.1. *The sum of diagonal entries of M is always even, since it equals $2G$ (twice the total number of games G), as each game contributes $+1$ to exactly two diagonal entries.*

Fixing Singularity: The Sum Constraint

Since M is singular, modify as follows:

1. Replace the **last row** of M with all 1s.
2. Replace the **last entry** of $\bar{q}$ with 0.

This enforces $r_1 + \cdots + r_T = 0$ (rankings centered at zero). The modified system is invertible and has a unique solution.

4.3 Shortcut: Building $M\hat{\mathbf{r}} = \bar{q}$ Directly

Rather than assembling A first:

- Set m_{ii} = games played by team i (for $i < T$).
- Set $m_{ij} = -($ games between i and $j)$ for $i \neq j$ (for $i < T$).
- Last row: all 1s.
- Set q_i = total points won by team i (for $i < T$).
- Last entry of $\bar{q}$: 0.

4.4 Worked Examples

Example 4.4.1 (Four Teams)**.** Games: T1–T2 (1 game), T1–T3 (1), T2–T3 (2), T2–T4 (1), T3–T4 (1). Results: T1 beats T2 by 7, T1 loses to T3 by 8, T2 ties T3, T2 beats T3 by 15, T2 beats T4 by 5, T3 loses to T4 by 1.

Degrees: T1 plays 2, T2 plays 4, T3 plays 4, T4 plays 2. Point totals: $q_1 = 7 - 8 = -1$, $q_2 = -7 + 0 + 15 + 5 = 13$, $q_3 = 8 + 0 - 15 - 1 = -8$, $q_4 = -5 + 1 = -4$.

$$M = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 4 & -2 & -1 \\ -1 & -2 & 4 & -1 \\ 1 & 1 & 1 & 1 \end{pmatrix}, \quad \bar{q} = \begin{pmatrix} -1 \\ 13 \\ -8 \\ 0 \end{pmatrix}.$$

Solving: $\hat{r}_1 \approx 0.125$, $\hat{r}_2 \approx 2.375$, $\hat{r}_3 \approx -1.125$, $\hat{r}_4 \approx -1.375$.

Order: **T2** > **T1** > **T3** > **T4**.

Prediction T1 vs T4 (never played): $r_1 - r_4 = 1.5$ (T1 expected to win by 1.5).

4.5 Special Situations

Symmetric Round Robin

If T1 beats T2 by k, T2 beats T3 by k, T3 beats T1 by k: the point totals are all zero ($q_1 = k - k = 0$, etc.), giving all equal rankings $r_1 = r_2 = r_3 = 0$. The cyclic symmetry of the results implies no team has a net advantage.

Effect of Repetition

If T1 beats T2 by n (once) vs. beats T2 by 1 (n times):

- **Once by n:** $r_1 = (n-1)/3$ (grows without bound as $n \to \infty$).
- **n times by 1:** $r_1 = (n-1)/(2n+1) \to 1/2$ (bounded).

One large win counts more than many small wins.

Home Advantage

Reduce winning margins for home victories by a fixed percentage (e.g., 25%) before entering the margins into the Massey system.

Disconnected Team Graphs

If the teams split into two groups with no games between groups, the Massey matrix (before modification) is block-singular and cannot be solved globally. Apply Massey separately to each connected group.

4.6 Prediction Formula

If teams i and j have never played, the expected margin when they meet is:

$$\hat{p}_{ij} = r_i - r_j.$$

A positive value means team i is expected to win.

Chapter 5

Graph Theory and the Fiedler Method

5.1 Graph Basics

Definition 5.1.1 (Graph, Simple Graph, Connected)**.** A **graph** $G = (V, E)$ is a set of **vertices** (nodes) V and **edges** E (pairs of vertices). A graph is:

- **Simple**: no loops (edges from a vertex to itself) and no multiple edges.
- **Connected**: there is a path between every pair of vertices.
- k**-regular**: every vertex has exactly k neighbors (degree k).

5.2 Matrix Representations

Definition 5.2.1 (Adjacency Matrix)**.** The **adjacency matrix** A of a simple graph on n vertices is the $n \times n$ symmetric matrix with $a_{ij} = 1$ if $\{i, j\} \in E$ and $a_{ij} = 0$ otherwise.

Definition 5.2.2 (Degree Matrix and Laplacian)**.**

$$D = \operatorname{diag}(\deg v_1, \ldots, \deg v_n), \qquad L = D - A.$$

L is the **Laplacian matrix**.

Example 5.2.1. For the path graph $1 - 2 - 3$:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

5.3 Walks and Powers of A

Definition 5.3.1 (Walk)**.** A **walk** of length k from vertex i to vertex j is a sequence of k edges starting at i and ending at j (vertices and edges may repeat).

Theorem 5.3.1 (Walks via A^k)**.** $(A^k)_{ij}$ = *number of walks of length k from vertex i to vertex j.*

Proof. Induction on k. Base: $(A^1)_{ij} = a_{ij}$ = number of walks of length 1 (0 or 1). Step: $(A^{k+1})_{ij} = \sum_\ell (A^k)_{i\ell} a_{\ell j}$ sums walks of length k to ℓ followed by one edge $\ell \to j$. □

Definition 5.3.2 (Trace)**.** $\operatorname{tr}(M) = \sum_i m_{ii}$.

Theorem 5.3.2 (Triangle Count)**.** *Number of triangles* $= \dfrac{1}{6}\operatorname{tr}(A^3)$.

Proof. A closed walk of length 3 from v to v in a simple graph must traverse exactly one triangle. Each triangle has 3 vertices and can be traversed in 2 directions, contributing 2 to each of the 3 diagonal entries. So $\operatorname{tr}(A^3) = 6 \times (\text{triangles})$. □

Remark 5.3.1 (Why Not $\operatorname{tr}(A^4)$ for Squares?). Closed walks of length 4 include both squares *and* backtracks ($v \to u \to v \to u \to v$) and triangle-backtracks. Unlike triangles, not all closed 4-walks are squares, so $\operatorname{tr}(A^4)$ overcounts. The triangle formula works because the only closed walk of length 3 in a simple graph *is* a triangle.

5.4 Graph Isomorphism

Definition 5.4.1 (Isomorphism)**.** Graphs G_1 and G_2 (both on n vertices) are **isomorphic** ($G_1 \cong G_2$) if there exists a permutation matrix P such that $PA_1P^T = A_2$, i.e., the vertices can be relabeled to make the adjacency matrices equal.

Theorem 5.4.1 (Triangle Count Invariant)**.** *If* $G_1 \cong G_2$ *then* $\frac{1}{6}\operatorname{tr}(A_1^3) = \frac{1}{6}\operatorname{tr}(A_2^3)$.

Example 5.4.1 (Non-Isomorphic Graphs)**.** The triangle K_3 has adjacency matrix $\begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ (1 triangle), while the path P_3 has $\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$ (0 triangles). They are not isomorphic.

Remark 5.4.1. The converse does not hold: two non-isomorphic graphs can have the same triangle count. For example, K_4 and two disjoint K_3 triangles both have 4 triangles but are not isomorphic (K_4 is connected, $K_3 \sqcup K_3$ is not).

5.5 Graph Partitioning

Definition 5.5.1 (Partition and Cut)**.** A **2-partition** (V_1, V_2) of V divides the vertices into two nonempty disjoint sets. The **cut** is the number of edges between V_1 and V_2.

Goal: Find a balanced partition (roughly $|V_1| \approx |V_2|$) with minimum cut.

Lemma 5.5.1. *Assign* $x_i = +1$ *for* $i \in V_1$ *and* $x_i = -1$ *for* $i \in V_2$. *Then* $\operatorname{cut}(V_1, V_2) = \frac{1}{4}\sum_{\{i,j\}\in E}(x_i - x_j)^2$.

5.6 The Fiedler Method

Fact 5.6.1 (Laplacian Eigenvalues)**.** *For a simple connected graph on n vertices, the eigenvalues of L satisfy:*

$$0 = \lambda_1 < \lambda_2 \le \lambda_3 \le \cdots \le \lambda_n.$$

The smallest eigenvalue is always 0 (eigenvector $\mathbf{1} = (1, 1, \ldots, 1)^T$*). The graph is connected* $\iff \lambda_2 > 0$.

Definition 5.6.1 (Fiedler Value and Fiedler Vector)**.**

- The **Fiedler value** (algebraic connectivity) is λ_2.
- A **Fiedler vector** is any eigenvector corresponding to λ_2.

Theorem 5.6.1 (Fiedler Partitioning Method)**.** *Compute the Fiedler vector* $\mathbf{v}$*. Partition:*

$$V_+ = \{i : v_i > 0\}, \qquad V_- = \{i : v_i < 0\}.$$

Vertices with $v_i = 0$ *may go in either group.*

Algorithm:

1. Build $L = D - A$.
2. Compute eigenpairs of L; sort eigenvalues.
3. The eigenvector for the second smallest eigenvalue is the Fiedler vector.
4. Partition by sign.

Example 5.6.1 (Triangle with Pendant)**.** Graph: 1–2, 1–3, 2–3, 3–4.

$$L = \begin{pmatrix} 2 & -1 & -1 & 0 \\ -1 & 2 & -1 & 0 \\ -1 & -1 & 3 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}.$$

Eigenvalues: $0, 1, 3, 4$. Fiedler vector: $\approx (-0.408, -0.408, 0, 0.816)^T$.

Partition: $V_- = \{1, 2\}$ (negative), $V_0 = \{3\}$ (zero, bridge vertex), $V_+ = \{4\}$ (positive). This correctly identifies the triangle group vs. the pendant vertex.

Example 5.6.2 (Complete Graph K_n)**.** The Laplacian of K_n is $L = nI - J$ where J is the all-ones matrix. Eigenvalues: 0 (once) and n ($n - 1$ times). So $\lambda_2 = n$ and the Fiedler value equals the number of vertices.

Remark 5.6.1 (Useful Facts about the Fiedler Vector).

- Since L is symmetric, it has real eigenvalues and an orthonormal eigenbasis.
- The Fiedler vector is orthogonal to $\mathbf{1}$, so its entries sum to 0.
- The partition produced approximately minimizes cut while balancing group sizes.
- If the eigenspace for λ_2 has dimension > 1, multiple Fiedler vectors exist; each can give a (potentially different but equally valid) partition.

Chapter 6

Markov Chains

This chapter covers Markov chains from two sources: (1) Justin Wyss-Gallifent's notes (Ch. 6), and (2) Lay & McDonald, *Linear Algebra and Its Applications*, Chapter 10, Sections 10.1–10.3.

6.1 Introduction and Motivation

Two-State Example

Suppose a total population of 1 is split between two states: 0.7 in State 1 and 0.3 in State 2. Each year, 10% of State 1 moves to State 2, and 5% of State 2 moves to State 1. After one year:

$$\text{State 1} = 0.90(0.7) + 0.05(0.3) = 0.645, \qquad \text{State 2} = 0.10(0.7) + 0.95(0.3) = 0.355.$$

In matrix form with transition matrix T:

$$T = \begin{pmatrix} 0.90 & 0.05 \\ 0.10 & 0.95 \end{pmatrix}, \quad \mathbf{x}_1 = T\mathbf{x}_0, \quad \mathbf{x}_n = T^n\mathbf{x}_0.$$

Column convention: Column j of T is an "exit vector" from state j. Entry T_{ij} = fraction moving from state j to state i in one step.

Lay-McDonald 10.1: Finite-State Markov Chains

A **finite-state Markov chain** is a mathematical model satisfying:

1. The next state depends **only** on the current state (Markov property – no memory).
2. Transition probabilities do not change with time (time-homogeneous).

Applications include signal transmission, gas diffusion (Ehrenfest model), and random walks.

6.2 Ehrenfest Diffusion Model (Lay-McDonald 10.1)

Setup: Two urns (A and B) containing $2k$ molecules total. At each step, pick one molecule uniformly at random and move it to the other urn. Track the number of molecules in urn A.

For $k = 3$ (6 molecules), states are $0, 1, \ldots, 6$. Transition probabilities from state i:

$$P(i \to i+1) = \frac{6-i}{6}, \qquad P(i \to i-1) = \frac{i}{6}.$$

States 0 and 6 transition deterministically to 1 and 5 respectively. The resulting 7×7 transition matrix has a checkered structure.

Remark 6.2.1. The Ehrenfest model does not reach a unique steady state because it is not regular (the transition matrix alternates between two classes of states). Long-term behavior depends on the parity of the initial state.

6.3 Random Walks (Lay-McDonald 10.1)

Random walk on $\{1, 2, \ldots, n\}$: At each step, move left with probability p or right with probability $1 - p$.

- **Absorbing boundaries:** States 1 and n are absorbing (once reached, stay there).
- **Reflecting boundaries:** States 1 and n bounce back inward.

Transition matrices:

$$P_{\text{absorbing}} = \begin{pmatrix} 1 & p & 0 & \cdots \\ 0 & 0 & p & \cdots \\ 0 & 1-p & 0 & \cdots \\ \vdots & & & \ddots \end{pmatrix}, \qquad P_{\text{reflecting}} = \begin{pmatrix} 0 & p & 0 & \cdots \\ 1 & 0 & p & \cdots \\ 0 & 1-p & 0 & \cdots \\ \vdots & & & \ddots \end{pmatrix}.$$

Random walk on a directed graph: States are vertices; at each step, follow a uniformly random outgoing edge. This is the basis for Google PageRank.

6.4 Steady States and Limits

Definition 6.4.1 (Probability Vector)**.** A vector $\mathbf{x}$ with non-negative entries summing to 1.

Definition 6.4.2 (Transition Matrix (Stochastic Matrix))**.** A matrix T whose columns are each probability vectors. Entry T_{ij} = fraction of population moving from state j to state i per step.

Definition 6.4.3 (Steady-State Vector)**.** A probability vector $\mathbf{x}^*$ with $T\mathbf{x}^* = \mathbf{x}^*$, i.e., an eigenvector of T for eigenvalue $\lambda = 1$.

Definition 6.4.4 (Regular Transition Matrix)**.** T is **regular** if T^k has all positive entries for some $k \geq 1$.

Theorem 6.4.1 (Convergence – Wyss-Gallifent & Lay-McDonald 10.2)**.** *If T is a regular transition matrix, then:*

1. *T has a **unique** steady-state vector $\mathbf{x}^*$.*
2. *For any probability vector $\mathbf{x}_0$: $\lim_{k\to\infty} T^k \mathbf{x}_0 = \mathbf{x}^*$.*
3. *$\lim_{k\to\infty} T^k$ exists and has all columns equal to $\mathbf{x}^*$.*

Proof of (3) given (2). Let $\mathbf{e}_i$ be the i-th standard basis vector. By (2), $\lim_{k\to\infty} T^k \mathbf{e}_i = \mathbf{x}^*$ for each i. Hence:

$$\lim_{k\to\infty} T^k = \lim_{k\to\infty} T^k [\mathbf{e}_1 \ \cdots \ \mathbf{e}_n] = [\mathbf{x}^* \ \cdots \ \mathbf{x}^*]. \quad \square$$

$\square$

Remark 6.4.1 (Non-Regular Matrices)*.* A non-regular T may have multiple steady-state vectors or no convergence. Examples:

- $T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$: oscillates forever, does not converge except from $\mathbf{x}_0 = (0.5, 0.5)^T$.
- A block upper-triangular T: population drains into absorbing bottom-right block.

Meaning of T_{ij}^k

The (i, j)-entry of T^k equals the fraction of the population starting in state j that is in state i after exactly k steps. This follows from the matrix product formula for T^k.

6.5 Finding the Steady State

Method 1 (Eigenvalue): Solve $(T - I)\mathbf{x} = \mathbf{0}$ with $\sum x_i = 1$. Software gives a unit eigenvector; divide by its entry-sum to get the probability vector.

Method 2 (Iteration): Compute $T^k\mathbf{x}_0$ for large k; the result converges to $\mathbf{x}^*$ by Theorem 6.4.1.

Method 3 (Column of T^k): Pick any column of T^k for large k; it converges to $\mathbf{x}^*$.

Example 6.5.1 (Three States)**.** $T = \begin{pmatrix} 0.90 & 0.10 & 0.20 \\ 0.06 & 0.80 & 0.10 \\ 0.04 & 0.10 & 0.70 \end{pmatrix}$. Eigenvalues: $1, 0.763, 0.637$. Eigenvector for $\lambda = 1$ (normalized): $\mathbf{x}^* \approx (0.581, 0.256, 0.163)^T$.

6.6 Proof for the 2×2 Case

Theorem 6.6.1. *If* $T = \begin{pmatrix} a & 1-b \\ 1-a & b \end{pmatrix}$ *is a regular transition matrix* ($0 < a, b < 1$, $(a, b) \neq (0, 0), (1, 1)$)*, then* $\lim_{n\to\infty} T^n\mathbf{x}_0 = \mathbf{x}^*$ *for all probability vectors* $\mathbf{x}_0$*.*

Proof. The eigenpairs of T are:

$$\left(\lambda_1 = 1,\ \mathbf{v}_1 = \begin{pmatrix} b-1 \\ a-1 \end{pmatrix}\right), \qquad \left(\lambda_2 = a + b - 1,\ \mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}\right).$$

Regularity forces $-1 < \lambda_2 = a + b - 1 < 1$, so $\lambda_2^n \to 0$. Diagonalizing:

$$T^n = P \begin{pmatrix} 1 & 0 \\ 0 & \lambda_2^n \end{pmatrix} P^{-1} \xrightarrow{n\to\infty} P \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} P^{-1} = \begin{pmatrix} \frac{b-1}{a+b-2} & \frac{b-1}{a+b-2} \\ \frac{a-1}{a+b-2} & \frac{a-1}{a+b-2} \end{pmatrix}.$$

Multiplying by any probability vector $\mathbf{x}_0 = \begin{pmatrix} c \\ 1-c \end{pmatrix}$ yields $\begin{pmatrix} \frac{b-1}{a+b-2} \\ \frac{a-1}{a+b-2} \end{pmatrix} = \mathbf{x}^*$ (independent of c).

□ □

6.7 Communication Classes (Lay-McDonald 10.3)

Definition 6.7.1 (Communicating States)**.** States i and j **communicate** (written $i \leftrightarrow j$) if the chain can reach state i from state j *and* state j from state i (in some number of steps). Communication is an equivalence relation, and its equivalence classes are called **communication classes**.

Definition 6.7.2 (Absorbing State)**.** State i is **absorbing** if $T_{ii} = 1$ (once entered, never left). Equivalently, state i forms its own communication class and has no outgoing transitions to other states.

Definition 6.7.3 (Absorbing Markov Chain)**.** A Markov chain is an **absorbing Markov chain** if:

1. It has at least one absorbing state, and
2. From every non-absorbing state, it is possible to reach some absorbing state.

Theorem 6.7.1 (Absorbing Chains Converge)**.** *In an absorbing Markov chain, the chain is absorbed with probability 1 (eventually reaches an absorbing state and stays there).*

Example 6.7.1 (Random Walk with Absorbing Boundaries)**.** For the random walk on $\{1, 2, 3, 4, 5\}$ with states 1 and 5 absorbing: states 2, 3, 4 are transient (eventually leave and never return). The chain is always eventually absorbed at state 1 or state 5. The probability of absorption at state 1 vs. state 5 depends on the starting state and the step probability p.

Remark 6.7.1 (Standard Form for Absorbing Chains). Reorder states so absorbing states come first. The transition matrix takes the standard form:

$$P = \begin{pmatrix} I & R \\ 0 & Q \end{pmatrix}$$

where I is the identity on absorbing states, Q is the transition matrix among transient states, and R is the matrix of transitions from transient to absorbing states.

Remark 6.7.2 (Non-Absorbing Non-Regular Chains). A chain can be neither regular nor absorbing. The Ehrenfest diffusion model is an example: it does not converge to a steady state from all starting points, yet it has no absorbing states. Such chains typically have periodic behavior.

6.8 Directed Graphs and Markov Chains (Lay-McDonald 10.1)

Definition 6.8.1 (Directed Graph)**.** A **directed graph** (digraph) has edges with direction: an edge from i to j means we can go from i to j but not necessarily from j to i.

The transition matrix for a random walk on a directed graph: if vertex j has out-degree d_j, then $T_{ij} = 1/d_j$ if there is an edge $j \to i$, and $T_{ij} = 0$ otherwise.

Example 6.8.1 (Signal Transmission)**.** Each bit (0 or 1) passes through a noisy channel. With probability p, a bit passes correctly; with probability $1 - p$, it flips. The transition matrix is:

$$P = \begin{pmatrix} p & 1-p \\ 1-p & p \end{pmatrix}.$$

After two stages, the probability a 0 remains a 0 is $(P^2)_{00} = p^2 + (1-p)^2$. For $p = 0.99$: $(P^2)_{00} = 0.9802$.

Chapter 7

Google PageRank

7.1 The Random Websurfer Model

PageRank models a **Random Websurfer (RW)** navigating the internet:

1. The RW is at some page.
2. If the page has outbound links: with probability 0.85 the RW follows a uniformly random link; with probability 0.15 the RW jumps to a uniformly random page anywhere.
3. If the page has **no outbound links** (dangling node): the RW jumps uniformly to any page.

This is a Markov chain. The state space is the set of all webpages.

7.2 The PageRank Transition Matrix (Lay-McDonald 10.2)

Definition 7.2.1 (Link Matrix S)**.** Let n be the number of pages. The **link matrix** S is $n \times n$ with:

$$S_{ij} = \begin{cases} 1/k & \text{page } j \text{ links to page } i \text{ and } j \text{ has } k \text{ outbound links,} \\ 1/n & \text{page } j \text{ has no outbound links (dangling node).} \end{cases}$$

Definition 7.2.2 (Google Matrix / PageRank Transition Matrix)**.**

$$T = \frac{0.15}{n}\mathbf{1}\mathbf{1}^T + 0.85\,S,$$

where $\mathbf{1}\mathbf{1}^T$ is the $n \times n$ all-ones matrix J.

Theorem 7.2.1. *T is a regular transition matrix (all entries $\geq 0.15/n > 0$, so T^1 has all positive entries).*

By Theorem 6.4.1 (the Convergence Theorem for regular Markov chains), T has a unique steady-state vector $\mathbf{x}^*$, which is the **PageRank vector**.

Definition 7.2.3 (PageRank)**.** The **PageRank** of page i is the i-th entry of the steady-state vector $\mathbf{x}^*$ satisfying $T\mathbf{x}^* = \mathbf{x}^*$, $\sum x_i^* = 1$.

7.3 Interpretation

The PageRank of page i equals the long-run fraction of time the Random Websurfer spends at page i. Pages with higher PageRank are considered more important.

Why this is better than counting inbound links:

- Being linked by an important page contributes more than being linked by a trivial one.
- Importance is defined *recursively*: a page is important if important pages link to it.
- This self-referential definition is captured by $T\mathbf{x}^* = \mathbf{x}^*$ (eigenvector equation).

Remark 7.3.1 (Outbound Links). Outbound links affect where traffic *leaves* your page, not where it arrives. Adding or removing outbound links does not change your PageRank.

7.4 Computing PageRank

Eigenvector method: Solve $T\mathbf{x} = \mathbf{x}$ (i.e., $(T - I)\mathbf{x} = \mathbf{0}$) with the constraint $\sum x_i = 1$. Since T is regular, $\lambda = 1$ is the unique eigenvalue of maximum modulus.

Power iteration: Start with any probability vector $\mathbf{x}_0$ (e.g., $\mathbf{x}_0 = (1/n, \ldots, 1/n)^T$) and compute $\mathbf{x}_{k+1} = T\mathbf{x}_k$. Since T is regular, this converges to $\mathbf{x}^*$.

Scalability note: For the actual web ($n \sim 10^{10}$), we do not compute eigenvalues of T (an $n \times n$ characteristic polynomial). Power iteration only requires matrix-vector products, which is far more tractable.

7.5 Special Cases and Examples

Example 7.5.1 (Two Symmetric Pages)**.** P1 $\leftrightarrow$ P2. By symmetry, both pages have PageRank 0.5.

Example 7.5.2 (Fully Connected Internet)**.** All n pages link to all $n - 1$ others. The uniform vector $\mathbf{u} = (1/n, \ldots, 1/n)^T$ satisfies $T\mathbf{u} = \mathbf{u}$. All pages have equal PageRank $1/n$.

Example 7.5.3 (Dangling Node)**.** Page n has no outbound links. Its column in S is $(1/n, \ldots, 1/n)^T$ (uniform random jump). The 0.15 random-jump term already covers 15% of this; the remaining 85% becomes the dangling contribution.

Example 7.5.4 (Disconnected Internet)**.** Without the random-jump term (i.e., $d = 1$), a disconnected internet may have no regular transition matrix and no unique steady state. The random jump ($d = 0.85$) adds $0.15/n > 0$ to every entry of T, making it regular and ensuring convergence.

7.6 Effect of the Damping Factor

The damping factor $d = 0.85$ (probability of following a link vs. jumping randomly) was chosen by Brin and Page empirically. Changing d generally preserves the ranking order but affects the magnitudes of PageRanks. As $d \to 1$, the ranking becomes more sensitive to link structure; as $d \to 0$, it approaches uniform $1/n$.

7.7 Chain Internet

For the chain $P_1 \to P_2 \to \cdots \to P_n$: page P_n is a dangling node (no outbound links), contributing random jumps to all pages. The transition matrix is:

$$T = \frac{0.15}{n} J + 0.85\, S, \quad S_{ij} = \begin{cases} 1 & j \to i \text{ is an edge, } j < n \\ 1/n & j = n \end{cases}.$$

Chapter 8

Spectral Theorem

Source: Lay & McDonald, Linear Algebra and Its Applications, §7.1.

Remark 8.0.1 (Suggested Problems 4). **Quiz:** Lay §7.1.13–22 (orthogonal diagonalization). **Exam:** Lay §7.1.25–38 (true/false and proofs).

8.1 Symmetric Matrices and Orthogonal Diagonalization

Definition 8.1.1 (Symmetric Matrix)**.** A matrix A is **symmetric** if $A^T = A$. Symmetric matrices are square.

Theorem 8.1.1 (Eigenvectors of Symmetric Matrices)**.** *If A is symmetric, then any two eigenvectors from different eigenspaces are orthogonal.*

Definition 8.1.2 (Orthogonally Diagonalizable)**.** An $n \times n$ matrix A is **orthogonally diagonalizable** if there exist an orthogonal matrix P and a diagonal matrix D such that $A = PDP^T = PDP^{-1}$.

Theorem 8.1.2 (Spectral Theorem)**.** *An $n \times n$ matrix A is orthogonally diagonalizable* if and only if *A is symmetric. Equivalently: A has n real eigenvalues (counting multiplicities); the dimension of each eigenspace equals the multiplicity of the corresponding eigenvalue; eigenspaces for different eigenvalues are mutually orthogonal.*

8.2 Spectral Decomposition

If $A = PDP^T$ with orthonormal eigenvectors $u_1, \ldots, u_n$ and eigenvalues $\lambda_1, \ldots, \lambda_n$, then

$$A = \lambda_1 u_1 u_1^T + \lambda_2 u_2 u_2^T + \cdots + \lambda_n u_n u_n^T.$$

Each term $\lambda_i u_i u_i^T$ is a rank-1 matrix; $u_i u_i^T$ is the orthogonal projection onto $\operatorname{span}\{u_i\}$.

8.3 Repeated Eigenvalues

When an eigenvalue has multiplicity > 1, the eigenspace has dimension equal to the multiplicity. Use Gram–Schmidt on a basis of the eigenspace to obtain orthonormal eigenvectors for the orthogonal diagonalization.

Chapter 9

Singular Value Decomposition

Sources: Justin Wyss-Gallifent, Singular Value Decomposition; Lay & McDonald, Linear Algebra and Its Applications, §7.4. The Spectral Theorem (Ch. 7.1) is used to construct the SVD.

Remark 9.0.1 (Suggested Problems 4). **Quiz:** Lay §7.4.5–12; Justin 8.1. **Exam:** Lay §7.4.13–18.

9.1 Introduction

Factoring a matrix means writing it as a product of other matrices. For example, if A is diagonalizable we can write $A = PDP^{-1}$, where D contains eigenvalues and P contains eigenvectors.

The **singular value decomposition (SVD)** is another factorization used in data compression, matrix approximation, pseudoinverses, signal analysis, and recognition.

9.2 Definitions

Definition 9.2.1 (Singular Value Decomposition)**.** For any $m \times n$ real matrix A, the **singular value decomposition** of A is a factorization

$$A = U\Sigma V^T$$

where:

- U is an $m \times m$ orthogonal matrix.
- Σ is an $m \times n$ rectangular diagonal matrix with diagonal entries $\sigma_1, \ldots, \sigma_{\min(m,n)}$ all nonnegative and in nonincreasing order.
- V is an $n \times n$ orthogonal matrix.

Definition 9.2.2 (Orthogonal Matrix)**.** An $n \times n$ matrix U is **orthogonal** if its columns are unit vectors all perpendicular to one another. Equivalently: columns form an orthonormal basis of $\mathbb{R}^n$, and $U^TU = UU^T = I$.

Definition 9.2.3 (Rectangular Diagonal Matrix)**.** An $m \times n$ **rectangular diagonal** matrix has nonzero entries only in positions $(1, 1)$, $(2, 2)$, etc. Extra rows or columns (if $m \neq n$) are all zeros.

9.3 Constructing the SVD

9.3.1 Key Observations

Observation 1. If $k \leq \min(m, n)$ singular values are nonzero, then $A = U\Sigma V^T$ depends only on the first k columns of U and V. Moreover:

$$A\bar{v}_i = \sigma_i \bar{u}_i, \qquad A^T\bar{u}_i = \sigma_i \bar{v}_i \quad \text{for } 1 \leq i \leq k.$$

Observation 2. Consider AA^T (size $m \times m$) and A^TA (size $n \times n$):

- Both are symmetric; by the Spectral Theorem they have orthogonal eigenvectors.
- All eigenvalues are nonnegative: $\|A\bar{v}\|^2 = \bar{v}^T A^T A\bar{v} = \lambda\|\bar{v}\|^2 = \lambda$.
- There is a bijection between eigenpairs of AA^T and A^TA for nonzero eigenvalues:

$$(\sigma^2, \bar{u}) \mapsto \left(\sigma^2, \tfrac{1}{\sigma}A^T\bar{u}\right), \qquad (\sigma^2, \bar{v}) \mapsto \left(\sigma^2, \tfrac{1}{\sigma}A\bar{v}\right).$$

Observation 3. For each positive eigenvalue σ_i^2 common to both, we have paired unit eigenvectors $\bar{u}_i$ (left singular vector) and $\bar{v}_i$ (right singular vector) satisfying $A\bar{v}_i = \sigma_i\bar{u}_i$ and $A^T\bar{u}_i = \sigma_i\bar{v}_i$.

9.3.2 Construction in Brief

1. Find the positive eigenvalues of AA^T (or A^TA). Denote them $\sigma_1^2 \geq \sigma_2^2 \geq \cdots \geq \sigma_k^2$.
2. For each σ_i^2, find unit eigenvector $\bar{u}_i$ of AA^T and set $\bar{v}_i = \frac{1}{\sigma_i}A^T\bar{u}_i$ (or vice versa). Handle multiplicity by spanning eigenspaces.
3. Build Σ with σ_i on the diagonal; build U from $\bar{u}_i$ and V from $\bar{v}_i$. Fill remaining columns of U and V with eigenvectors for eigenvalue 0 to complete orthonormal bases.

9.4 Best Low-Rank Approximation

Theorem 9.4.1 (Eckart–Young)**.** *Among all matrices of rank $\leq k$, the best approximation to A (in Frobenius norm) that preserves the most variance is*

$$A_k = \sum_{i=1}^{k} \sigma_i\bar{u}_i\bar{v}_i^T = U\Sigma_k V^T,$$

where Σ_k is Σ with $\sigma_{k+1}, \ldots, \sigma_{\min(m,n)}$ set to zero.

The proportion of variance preserved is

$$\frac{\sigma_1^2 + \cdots + \sigma_k^2}{\sigma_1^2 + \cdots + \sigma_{\min(m,n)}^2}.$$

9.5 Fundamental Subspaces and Reduced SVD (Lay-McDonald §7.4)

Given $A = U\Sigma V^T$ with rank r: $\{u_1, \ldots, u_r\}$ is an orthonormal basis for $\operatorname{Col} A$; $\{u_{r+1}, \ldots, u_m\}$ for $\operatorname{Nul} A^T$; $\{v_1, \ldots, v_r\}$ for $\operatorname{Row} A$; $\{v_{r+1}, \ldots, v_n\}$ for $\operatorname{Nul} A$.

Reduced SVD: $A = U_r D V_r^T$ where U_r, V_r contain only the first r columns. The **pseudoinverse** is $A^+ = V_r D^{-1} U_r^T$; $\hat{x} = A^+b$ is the least-squares solution of $Ax = b$ with smallest norm.

Condition number: For invertible A, $\kappa(A) = \sigma_1/\sigma_n$ measures sensitivity of $Ax = b$ to perturbations.

Chapter 10

Single-Qubit Quantum Systems

Sources: Eleanor Rieffel and Wolfgang Polak, Quantum Computing: A Gentle Introduction, Chapter 2; Sheldon Axler, Linear Algebra Done Right, §1A ($\mathbb{R}^n$ and $\mathbb{C}^n$). Optional: John Watrous, Basics of Quantum Information (video lectures).

10.1 Complex numbers and $\mathbb{C}^n$ (Axler §1A)

Complex numbers. A complex number is $z = a + bi$ with $a, b \in \mathbb{R}$ and $i^2 = -1$. Addition and multiplication follow the usual rules with $i^2 = -1$ (Axler defines these explicitly). The set $\mathbb{C}$ is a field: commutativity, associativity, identities, inverses, and distributivity hold (Axler 1.3). The **complex conjugate** is $\overline{a + bi} = a - bi$; **modulus** $|z| = \sqrt{z\overline{z}}$.

Lists and $\mathbb{F}^n$. A **list** of length n is an ordered n-tuple $(z_1, \ldots, z_n)$. For $\mathbb{F}$ either $\mathbb{R}$ or $\mathbb{C}$, $\mathbb{F}^n$ is the set of all such lists with entries in $\mathbb{F}$. Addition and scalar multiplication are componentwise. Vectors in $\mathbb{C}^n$ are columns $(z_1, \ldots, z_n)^T$ when using matrix notation.

Inner product on $\mathbb{C}^n$ (standard). For $u = (w_1, \ldots, w_n)$ and $v = (z_1, \ldots, z_n)$ in $\mathbb{C}^n$,

$$\langle u, v \rangle = w_1\overline{z_1} + \cdots + w_n\overline{z_n}.$$

conjugate-linear in the *second* slot and linear in the first (matching Rieffel's bra/ket usage below). The **norm** is $\|v\| = \sqrt{\langle v, v \rangle}$.

Scalars and $\mathbb{F}^n$ (Axler). The letter $\mathbb{F}$ denotes either $\mathbb{R}$ or $\mathbb{C}$ (Axler 1.6). For fixed n, $\mathbb{F}^n$ is the set of all lists (ordered n-tuples) $(x_1, \ldots, x_n)$ with each $x_k \in \mathbb{F}$. Addition is coordinatewise; **scalar multiplication** is $\lambda(x_1, \ldots, x_n) = (\lambda x_1, \ldots, \lambda x_n)$ for $\lambda \in \mathbb{F}$. The zero vector is $0 = (0, \ldots, 0)$. These operations satisfy the vector-space axioms (proved in Axler from the field axioms for $\mathbb{F}$).

Fields (Axler "Digression"). Real and complex numbers are examples of **fields**: sets with addition and multiplication satisfying commutativity, associativity, identities, inverses (except 0 has no multiplicative inverse), and distributivity. All results stated for $\mathbb{F}^n$ apply with $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$.

Remark 10.1.1 (Inner-product conventions). Axler develops abstract inner products in Chapter 6; his complex convention may place the complex conjugate on the *first* slot rather than the second. Rieffel–Polak and these notes use $\langle \phi | \psi \rangle$ with conjugate-linearity in the bra—keep one convention fixed in every calculation.

10.2 Qubit as $\mathbb{C}^2$ (Rieffel–Polak §2.2)

The **state space** of a two-level quantum system is a two-dimensional complex inner-product space. A **qubit** is a *unit* vector in $\mathbb{C}^2$, representing a physical state up to a **global phase** $e^{i\varphi}$ (Rieffel §2.5): $|\psi\rangle$ and $e^{i\varphi}|\psi\rangle$ are the same state.

Fix an orthonormal **computational** (or **standard**) basis $\{|0\rangle, |1\rangle\}$. Any (pure) state is

$$|\psi\rangle = a\,|0\rangle + b\,|1\rangle\,, \qquad a, b \in \mathbb{C}, \quad |a|^2 + |b|^2 = 1.$$

The "two states" of a classical bit are $|0\rangle$ and $|1\rangle$; a qubit can be any superposition $a\,|0\rangle + b\,|1\rangle$ on the unit sphere in $\mathbb{C}^2$.

Linear algebra vocabulary (Rieffel §2.2). A ket is a **linear combination** of kets $|s_1\rangle, \ldots, |s_n\rangle$ if $|v\rangle = \sum_i a_i\,|s_i\rangle$ with $a_i \in \mathbb{C}$. The **span** of a set S consists of all such combinations. A **basis** is a linearly independent spanning set; in finite dimensions every basis has the same length (**dimension**). For qubits, dim = 2; any two non-collinear kets form a basis, but quantum mechanics usually uses **orthonormal** bases: $\langle\beta_i|\beta_j\rangle = \delta_{ij}$.

10.3 Dirac notation: bras, kets, inner product

A **ket** $|\psi\rangle$ is a column vector. The Hermitian conjugate (row) is the **bra** $\langle\psi| = |\psi\rangle^\dagger$. The **inner product** $\langle\phi|\psi\rangle$ is linear in $|\psi\rangle$ and conjugate-linear in $\langle\phi|$. Rieffel lists the axioms: $\langle\psi|\psi\rangle \geq 0$ real; $\langle\phi|\psi\rangle = \overline{\langle\psi|\phi\rangle}$; and linearity in the second argument. Orthogonal means $\langle\phi|\psi\rangle = 0$; **orthonormal** bases satisfy $\langle\beta_i|\beta_j\rangle = \delta_{ij}$.

Born rule: A projective measurement in orthonormal basis $\{|b_0\rangle, |b_1\rangle\}$ gives outcome k with probability

$$P(k) = |\,\langle b_k|\psi\rangle\,|^2,$$

and the post-measurement state is $|b_k\rangle$ (for an ideal projective measurement).

10.4 Photon polarization (Rieffel §2.1)

Physical model. Linear polarization of a photon is described by a unit vector in a real 2-dimensional plane (vertical $|\uparrow\rangle$, horizontal $|\rightarrow\rangle$, or an angle in between). More general polarization uses $\mathbb{C}^2$ (circular and elliptical polarization correspond to complex amplitudes).

Polarizer interaction (Born rule). A polarizer has a **preferred axis** (a unit ket $|u\rangle$). A photon in state $|\psi\rangle$ passes with probability $|\,\langle u|\psi\rangle\,|^2$ and is then measured to be in state $|u\rangle$; otherwise it is absorbed (orthogonal outcome).

Three-polarizer "paradox" (Rieffel §2.1.1–2.1.2). Two crossed polarizers (e.g. horizontal then vertical) transmit no light. Inserting a third polarizer at 45° between them allows some transmission: photons blocked classically can acquire amplitude in the vertical direction after passing the intermediate $|\diamondsuit\rangle$ (diagonal) polarizer, because the state is re-prepared in an intermediate basis before the final measurement.

10.5 Single-qubit measurement (Rieffel §2.3)

A **projective measurement** in an orthonormal basis $\{|b_0\rangle, |b_1\rangle\}$ yields outcome k with probability $|\,\langle b_k|\psi\rangle\,|^2$, and (for ideal measurement) collapses the state to $|b_k\rangle$. The same physical qubit can be measured in the computational basis $\{|0\rangle, |1\rangle\}$, the Hadamard basis $\{|+\rangle, |-\rangle\}$, the circular basis $\{|i\rangle, |-i\rangle\}$, or any other orthonormal basis; probabilities depend on the chosen basis.

10.6 BB84 quantum key distribution (Rieffel §2.4)

The **BB84** protocol lets two parties share a secret random bit string using qubits sent in randomly chosen bases (computational or Hadamard). Comparing a subset of bases and outcomes over a public channel reveals eavesdropping: an interceptor who measures in the wrong basis disturbs

the state. This uses the fact that non-orthogonal states cannot be distinguished reliably without disturbing them—a theme beyond the SP5 exercise list but central to Rieffel's §2.4.

10.7 State space: global phase, $\mathbb{C}\mathrm{P}^1$, Bloch sphere (Rieffel §2.5)

Unit vectors $|\psi\rangle$ and $e^{i\varphi}|\psi\rangle$ describe the **same** physical state (**global phase**). Observable phases are **relative** (e.g. the phase between amplitudes of $|0\rangle$ and $|1\rangle$ in a fixed basis). The space of distinct pure states of one qubit is the complex projective line $\mathbb{C}\mathrm{P}^1$ (quotient of the unit sphere in $\mathbb{C}^2$ by the global-phase equivalence). The **Bloch sphere** parametrizes $\mathbb{C}\mathrm{P}^1$ by a point on the ordinary sphere S^2, giving a geometric picture of qubit state space (Rieffel §2.5.2).

10.8 Named bases for one qubit

Computational: $\{|0\rangle, |1\rangle\}$.

Hadamard:

$$|+\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \qquad |-\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle).$$

Circular (y-axis) basis:

$$|i\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle + i\,|1\rangle), \qquad |-i\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle - i\,|1\rangle).$$

(Some texts denote circular polarization with separate symbols; always match the problem's definitions.)

A state is a **superposition** with respect to a chosen orthonormal basis if at least two basis coefficients are nonzero. The same physical state may be a superposition in one basis and a basis vector in another (e.g. $|+\rangle$ in $\{|0\rangle, |1\rangle\}$ vs. $\{|+\rangle, |-\rangle\}$).

10.9 Suggested Problems 5 (course alignment)

Suggested reading: Axler, *Linear Algebra Done Right*, §1A; Watrous, *Basics of Quantum Information* (videos); Rieffel–Polak, §2.
Quiz-level: Rieffel–Polak Exercises 2.2, 2.3, 2.4, 2.6.
Exam-level: N/A (similar difficulty to quiz-level).
Code: `ex9_quantum_qubit.m` (Octave checks for inner products and Born probabilities).

Remark 10.9.1 (What these notes cover). The sections above follow the *logical order* of ideas in Axler §1A and Rieffel §2; they are not a substitute for reading the books (examples, figures, proofs, and all exercises). **SP5 exercises solved in the solutions PDF:** Rieffel 2.2, 2.3, 2.4, 2.6 only (per the course handout). Rieffel's other Chapter 2 exercises (e.g. 2.1, 2.5, 2.7–2.14) are not assigned on SP5 but appear in the textbook for extra practice.

Chapter 11

Image Compression

Sources: Justin Wyss-Gallifent, Image Compression; Lay & McDonald, Linear Algebra and Its Applications, §7.5.

Remark 11.0.1 (Suggested Problems 4). **Quiz:** Justin 10.1–10.3. **Exam:** Justin 10.6. (Lay §7.5.1–6: practice for PCA/covariance.)

11.1 Image Representation

A grayscale image displayed pixel-by-pixel is stored as a matrix. Common conventions:

- **Color:** Red, Green, Blue each 0–255 (256^3 colors).
- **Grayscale (integer):** Single value 0–255 (0 = black, 255 = white).
- **Grayscale (real):** Single value in $[0, 1]$ (0 = black, 1 = white).

We use the third option (real values in $[0, 1]$) for mathematical convenience.

11.2 Image Compression via SVD

If a matrix A represents an image, then a low-rank approximation $A' = U\Sigma' V^T$ (with some singular values zeroed) approximates the image while preserving a fraction of the variance.

Handling out-of-range values: After reconstruction, entries may fall outside $[0, 1]$. Do *not* rescale (that would distort meaning). Instead: treat values > 1 as white and values < 0 as black.

Effect: As fewer singular values are kept, the image becomes more uniform while retaining as much variance as possible. Rank-1 images are a single vertical vector (shades of grey) scaled across columns.

11.3 Image Quality

Definition 11.3.1 (Image Quality)**.** The **image quality** (proportion of variance preserved) is

$$\frac{\sigma_1^2 + \cdots + \sigma_k^2}{\sigma_1^2 + \cdots + \sigma_{\min(m,n)}^2}.$$

A typical target is $\geq 99.5\%$ variance for a reasonable-looking picture.

11.4 Data Savings

For an $n \times n$ grayscale image:

- **Original:** n^2 values (one per pixel).
- **SVD with k singular values:** We store $\bar{u}_1, \ldots, \bar{u}_k$ (each n values) and the coefficients for each column (each column needs k coefficients, n columns $\Rightarrow$ kn values). Total: $kn + kn = 2kn$ values.

Definition 11.4.1 (Compression Ratio)**.**

$$\frac{2kn}{n^2} = \frac{2k}{n}.$$

Compression is beneficial when $2kn < n^2$, i.e., $k < n/2$.

For a general $m \times n$ (non-square) image, the same logic gives $2k(m+n)$ values if we store k left singular vectors (length m) and k right singular vectors (length n), plus the k singular values. The exact formula depends on the storage scheme; see Exercise 10.6.

11.5 Principal Component Analysis (Lay-McDonald §7.5)

For multivariate data $X_1, \ldots, X_N$ in $\mathbb{R}^p$, the **sample mean** is $M = \frac{1}{N}(X_1 + \cdots + X_N)$. Put data in **mean-deviation form**: $\hat{X}_k = X_k - M$. The **sample covariance matrix** is

$$S = \frac{1}{N-1} BB^T, \quad B = [\hat{X}_1 \ \cdots \ \hat{X}_N].$$

S is symmetric and positive semidefinite. The **principal components** are the unit eigenvectors of S ordered by decreasing eigenvalue. The first principal component u_1 gives the direction of maximum variance; $y_1 = u_1^T X$ has the largest variance among unit linear combinations. The fraction of total variance in the first component is $\lambda_1 / \operatorname{tr}(S)$.